

Few-Shot k-Nearest Neighbor Classification with Self-Supervised Visual Features

A frozen-backbone benchmark of seven SSL Vision Transformers on CIFAR-10 and CIFAR-100

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Project repository: <https://github.com/kostismatz/ssl-knn-fewshot>

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Abstract

This report evaluates the quality of visual representations learned by seven self-supervised learning (SSL) backbone models which are: DINOv1 ViT-S/16, DINOv2 ViT-S/14, DINOv2 ViT-S/14 with registers, DINOv3 ViT-S/16, MoCo v3 ViT-S/16, iBOT ViT-S/16 and AttMask ViT-S/16 by using a non-parametric k-nearest-neighbour (kNN) classifier as a probe. Features are extracted once from frozen, pretrained encoders and evaluated across a grid of few-shot regimes ($N \in \{1, 2, 5, 10, 20, 50, 100\}$ shots per class) on CIFAR-10 and CIFAR-100. Because the classifier has no trainable parameters, accuracy reflects purely the geometry of each backbone's representation space. The DINOv2 family is strongest (94.7% on CIFAR-10, 80.6% on CIFAR-100 at 100 shots), with DINOv3 a close second; iBOT and AttMask land between DINOv1 and DINOv2, while MoCo v3 trails all self-distillation methods by a wide margin.

1. Introduction

Self-supervised learning trains visual representations without human labels by solving pretext tasks derived from the data itself. A natural question is how semantically coherent these embeddings are without any task-specific adaptation. The k-nearest-neighbor classifier is an ideal probe: it has no learnable parameters, so any accuracy it achieves is attributable entirely to the structure of the pretrained feature space. In the few-shot regime this test is especially demanding, as the model must already separate class clusters well enough that a handful of labelled prototypes suffice. This work benchmarks seven ViT-Small backbones under this strict non-parametric protocol, varying the number of shots, the number of neighbors, and the feature pooling strategy.

2. Methodology

2.1 Backbone Models

All models are ViT-Small and are used as frozen feature extractors. Also no fine-tuning is performed at any stage.

Model	Architecture	SSL Objective	Pretrain Data
dino_vits16	ViT-S/16	Self-distillation	ImageNet-1K
dinov2_vits14	ViT-S/14	Self-dist. + MIM	LVD-142M
dinov2_vits14_reg	ViT-S/14 +reg	Self-dist. + MIM	LVD-142M
dinov3_vits16	ViT-S/16 +reg	Distilled (ViT-7B)	LVD-1689M
moco_v3_vits16	ViT-S/16	Momentum contrast	ImageNet-1K
ibot_vits16	ViT-S/16	Self-dist. + MIM	ImageNet-1K
attmask_vits16	ViT-S/16	Attention-guided MIM	ImageNet-1K

2.2 Feature Extraction

Raw 32×32 CIFAR images are processed entirely on GPU to maximize throughput: the dataset is loaded as a single tensor, normalized with ImageNet statistics, up sampled to 224×224 with bicubic interpolation, and passed through `model.forward_features()` under fp16 autocast. Two pooled descriptors are produced per image

- the CLS token

- the mean of the patch tokens (excluding CLS and any register tokens)

Both are L2-normalised so that cosine similarity reduces to a dot product. Features are cached to disk so the evaluation grid runs without recomputation.

2.3 k-Nearest-Neighbor Classifier

The classifier uses cosine similarity with DINO-style temperature-weighted voting: for each test sample the top-k support neighbors vote for their class with weight $\exp(\text{sim}/\tau)$, $\tau = 0.07$, accumulated per class with a vectorized scatter-add. The class with the highest score wins. k is swept over {1, 5, 20} and clamped to the support-set size at very low shots.

2.4 Few-Shot Sampling and Metrics

For each (model, dataset, pooling, N, k) combination, N examples per class are stratified-sampled from the training set; the full test set is always used for evaluation. Each setting is repeated over 3 random seeds and reported as mean $\pm$ standard deviation. Both Top-1 accuracy (primary) and macro-F1 (secondary) are computed.

3. Experimental Setup

The full grid spans 7 models $\times$ 2 datasets $\times$ 2 pooling modes $\times$ 7 shot counts $\times$ 3 k values $\times$ 3 seeds = 1,764 evaluated conditions. CIFAR-10 (10 classes) and CIFAR-100 (100 classes) each provide 50,000 train / 10,000 test images at 32 $\times$ 32 resolution. The temperature $\tau = 0.07$ is fixed throughout.

4. Results

All values below are mean % $\pm$ std % across 3 seeds at k = 5. The best value in each column is shown in bold.

4.1 CIFAR-10

CLS Token Pooling

Model	1-shot	5-shot	10-shot	100-shot
dino_vits16	53.2 $\pm$ 5.7	76.2 $\pm$ 2.0	80.1 $\pm$ 1.3	88.4 $\pm$ 0.5
dinov2_vits14	66.0 $\pm$ 8.0	87.0 $\pm$ 1.1	89.6 $\pm$ 0.4	94.7 $\pm$ 0.2
dinov2_vits14_reg	68.2 $\pm$ 5.8	87.8 $\pm$ 2.0	90.7 $\pm$ 0.8	94.6 $\pm$ 0.0
dinov3_vits16	56.5 $\pm$ 4.4	83.5 $\pm$ 1.1	86.8 $\pm$ 0.7	93.3 $\pm$ 0.3
moco_v3_vits16	21.8 $\pm$ 1.3	30.5 $\pm$ 1.5	33.5 $\pm$ 0.7	43.9 $\pm$ 0.5
ibot_vits16	57.6 $\pm$ 6.0	79.6 $\pm$ 0.8	83.1 $\pm$ 0.2	90.4 $\pm$ 0.4
attmask_vits16	56.7 $\pm$ 4.3	78.3 $\pm$ 1.6	82.0 $\pm$ 1.0	90.4 $\pm$ 0.4

Mean Pool Pooling

Model	1-shot	5-shot	10-shot	100-shot
dino_vits16	32.8 $\pm$ 5.1	46.6 $\pm$ 1.7	51.4 $\pm$ 1.2	64.4 $\pm$ 0.4
dinov2_vits14	66.2 $\pm$ 8.8	84.6 $\pm$ 0.2	86.4 $\pm$ 0.3	92.1 $\pm$ 0.2
dinov2_vits14_reg	69.6 $\pm$ 8.7	87.7 $\pm$ 2.0	90.1 $\pm$ 1.1	94.0 $\pm$ 0.3
dinov3_vits16	58.4 $\pm$ 7.4	80.9 $\pm$ 2.1	84.9 $\pm$ 0.6	90.4 $\pm$ 0.4
moco_v3_vits16	20.9 $\pm$ 0.7	31.2 $\pm$ 1.2	34.7 $\pm$ 0.9	43.8 $\pm$ 1.0
ibot_vits16	36.0 $\pm$ 5.4	58.0 $\pm$ 2.4	64.4 $\pm$ 1.2	76.3 $\pm$ 0.7
attmask_vits16	41.8 $\pm$ 4.7	59.0 $\pm$ 3.5	63.6 $\pm$ 1.4	75.5 $\pm$ 0.1

4.2 CIFAR-100

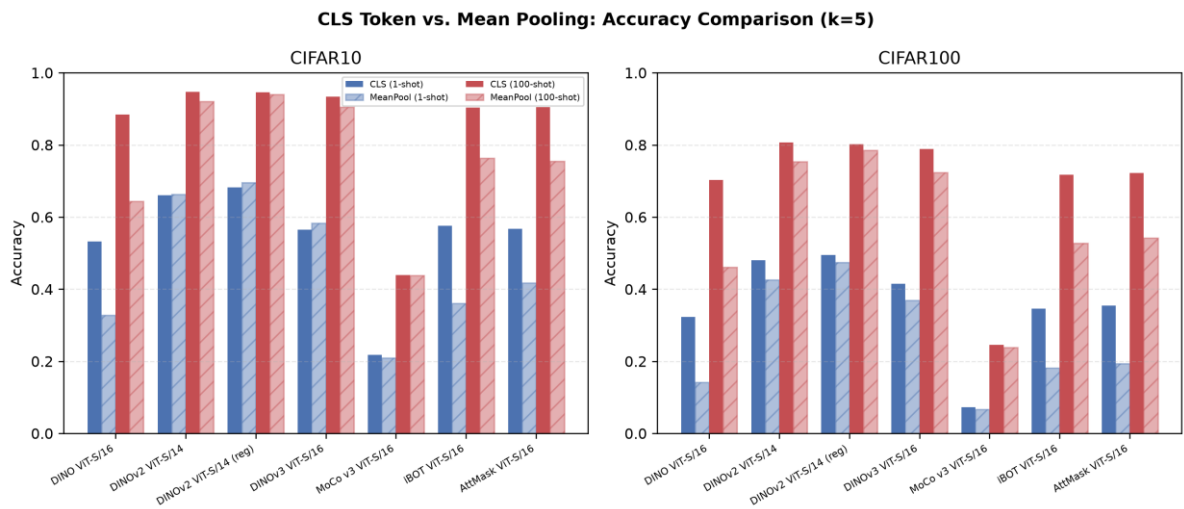
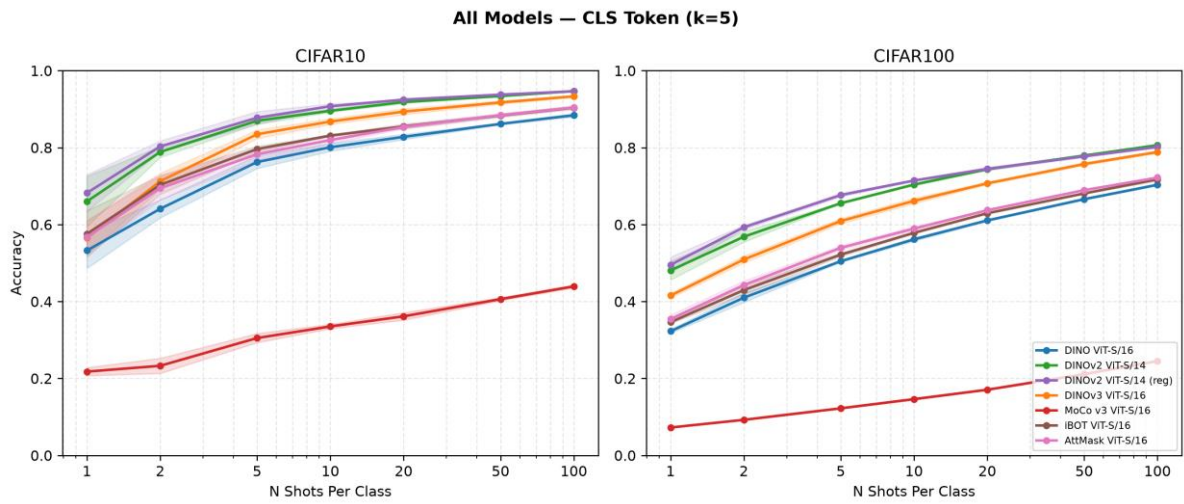
CLS Token Pooling

Model	1-shot	5-shot	10-shot	100-shot
dino_vits16	32.3 $\pm$ 0.6	50.5 $\pm$ 0.3	56.1 $\pm$ 0.6	70.3 $\pm$ 0.3
dinov2_vits14	48.1 $\pm$ 3.0	65.5 $\pm$ 0.2	70.4 $\pm$ 0.3	80.6 $\pm$ 0.3
dinov2_vits14_reg	49.5 $\pm$ 2.5	67.7 $\pm$ 0.5	71.4 $\pm$ 0.3	80.2 $\pm$ 0.3
dinov3_vits16	41.5 $\pm$ 0.6	60.9 $\pm$ 0.7	66.1 $\pm$ 0.7	78.8 $\pm$ 0.3
moco_v3_vits16	7.2 $\pm$ 0.0	12.2 $\pm$ 0.2	14.6 $\pm$ 0.2	24.5 $\pm$ 0.2
ibot_vits16	34.6 $\pm$ 0.4	52.2 $\pm$ 0.5	57.8 $\pm$ 0.3	71.7 $\pm$ 0.3
attmask_vits16	35.4 $\pm$ 1.0	53.9 $\pm$ 0.6	58.9 $\pm$ 0.5	72.2 $\pm$ 0.1

Mean Pool Pooling

Model	1-shot	5-shot	10-shot	100-shot
dino_vits16	14.1 $\pm$ 0.6	23.7 $\pm$ 0.6	28.8 $\pm$ 0.8	46.1 $\pm$ 0.5
dinov2_vits14	42.5 $\pm$ 2.5	58.5 $\pm$ 0.3	63.6 $\pm$ 0.3	75.3 $\pm$ 0.2
dinov2_vits14_reg	47.4 $\pm$ 1.3	63.9 $\pm$ 0.3	68.5 $\pm$ 0.2	78.5 $\pm$ 0.2
dinov3_vits16	36.9 $\pm$ 1.5	53.3 $\pm$ 0.4	58.5 $\pm$ 0.4	72.3 $\pm$ 0.1
moco_v3_vits16	6.6 $\pm$ 0.7	11.2 $\pm$ 0.6	13.2 $\pm$ 0.3	23.8 $\pm$ 0.2
ibot_vits16	18.2 $\pm$ 1.3	30.2 $\pm$ 0.3	36.1 $\pm$ 0.1	52.7 $\pm$ 0.2
attmask_vits16	19.3 $\pm$ 1.1	32.1 $\pm$ 0.3	37.4 $\pm$ 0.5	54.2 $\pm$ 0.2

4.3 Visual Analysis



Effect of k on kNN Accuracy (N=100 shots)

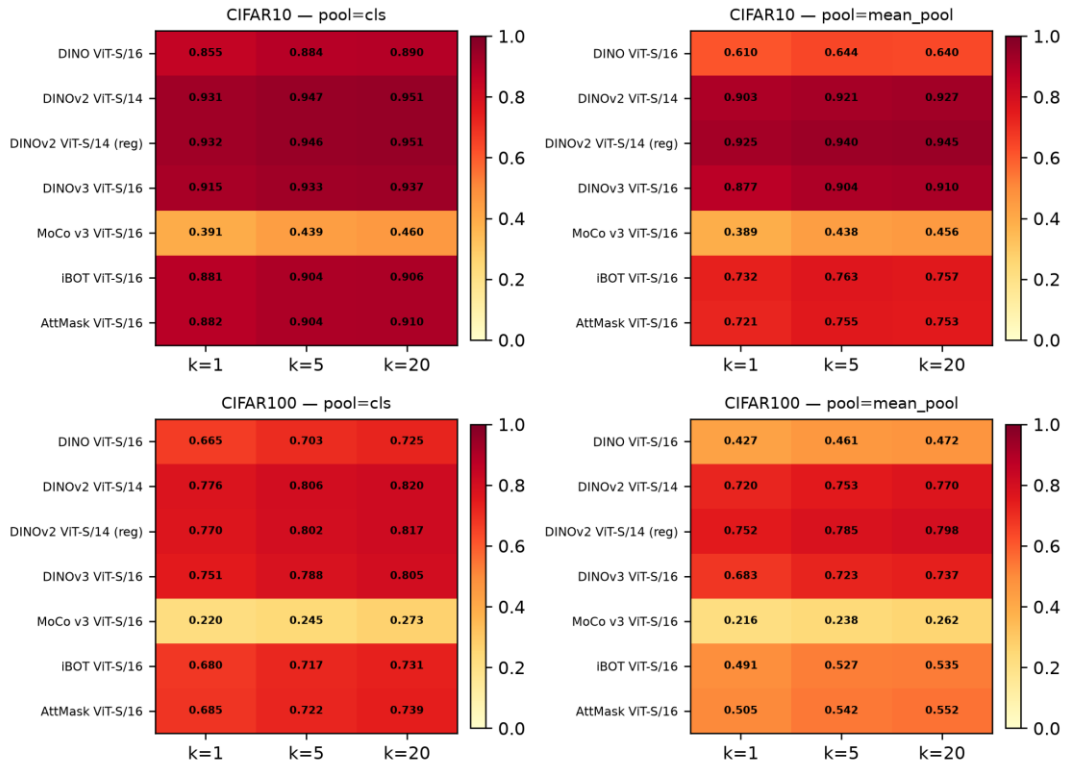


Figure 3. Effect of k on accuracy at N=100 shots.

Accuracy vs. Random-Chance Baseline (CLS, k=5)

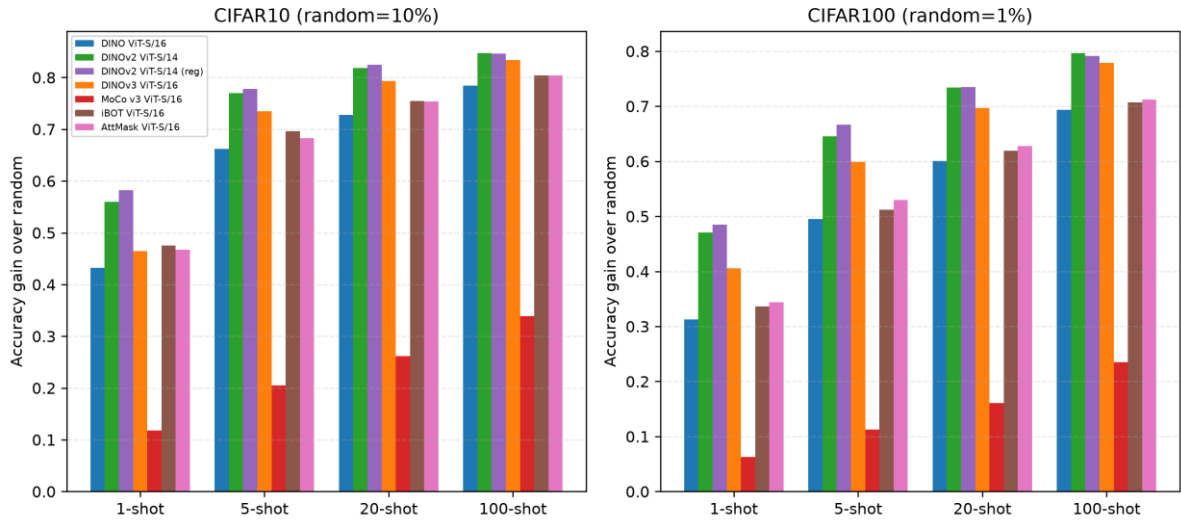


Figure 4. Accuracy gain over the random-chance baseline (CLS, k=5).

5. Discussion

5.1 DINOv2 and DINOv3 lead

The DINOv2 family is the strongest backbone at every shot count, reaching 94.7% on CIFAR-10 and 80.6% on CIFAR-100 at 100 shots. DINOv3 ViT-S/16, which is distilled from a giant ViT-7B teacher on the curated LVD-1689M corpus, is a close second (93.3% / 78.8%). The small residual edge of DINOv2 likely stems from its finer patch grid (ViT-S/14 yields 256 patches vs 196 for ViT-S/16) and its direct, non-distilled objective. Both keep mean-pool competitive with the CLS token, reflecting strong patch-level training.

5.2 Masked-image models: iBOT and AttMask

iBOT and AttMask combine self-distillation with masked image modelling and land squarely between DINOv1 and DINOv2 (90.4% on CIFAR-10, 71.7%/72.2% on CIFAR-100 at 100 shots). This ordering is intuitive: masked modelling enriches patch tokens, but both are pretrained on ImageNet-1K rather than the much larger curated corpus used by the DINOv2/v3 line. AttMask, which masks the most attention-salient patches, is marginally stronger than iBOT on CIFAR-100 and notably better under mean-pooling at low shots.

5.3 CLS token vs. mean pooling

The pooling effect depends on the backbone generation. For DINOv1 the CLS token is far superior (88.4% vs 64.4% at 100 shots on CIFAR-10), because its training optimizes the CLS token explicitly. For DINOv2/v3 the gap nearly closes, as masked image modelling makes individual patch tokens semantically rich. MoCo v3 is poor either way, indicating that representation quality, not the pooling choice, is its bottleneck.

5.4 MoCo v3 underperformance

MoCo v3 trails all DINO variants by 40+ percentage points. Its contrastive, instance-discrimination objective separates individual images but does not explicitly group semantically similar ones, which is exactly what kNN retrieval requires. Self-distillation methods, by contrast, learn semantic cluster structure directly.

5.5 Scaling and the effect of k

All models improve log-linearly with N, with the steepest gains between 1 and 10 shots and diminishing returns afterwards; variance across seeds collapses beyond ~20 shots. The number of neighbors k has a moderate effect ($k=1 \rightarrow k=20$ changes accuracy by 1–5 pp); DINOv2/v3 are robust to k, whereas weaker models such as MoCo v3 benefit more from larger k as voting smooths their noisier feature space.

5.6 Self-Supervised vs. Supervised Baseline

To put these numbers in context, four ResNet-50 networks were trained from scratch with full label supervision and probed with the identical kNN protocol, serving as a supervised upper bound (Appendix A). The comparison is striking: the best frozen self-supervised backbone matches or exceeds this supervised reference despite using no labels during pretraining. On CIFAR-10, DINOv2 reaches 94.7% versus the supervised ResNet-50's 94.6%; on CIFAR-100, DINOv2 (80.6%) actually surpasses the supervised baseline (76.6%) at 100 shots per class. In the accuracy-vs-shots curves the supervised ResNet appears as a near-flat ceiling that the strongest SSL models climb to meet which

concludes to the central evidence that modern self-supervised pretraining has effectively closed the gap to supervised feature learning on these benchmarks.

6. Supervised ResNet-50 Baseline

To contextualize the SSL results against traditional supervised learning, the project also trains ResNet-50 classifiers from scratch and evaluates them with the identical kNN protocol. Four variants are provided using CIFAR-10 and CIFAR-100, each with standard augmentation (random crop + horizontal flip) and a stronger augmentation regime (TrivialAugmentWide + random erasing). Their extracted features are mapped into the unified feature format via a conversion step so they can be scored within the same grid as the SSL backbones. This baseline provides a supervised reference point for the label-efficiency of the frozen SSL representations.

7. Conclusion

Frozen SSL ViT-Small backbones combined with a simple temperature-weighted kNN classifier achieve strong few-shot performance on CIFAR-10/100 without any gradient updates on the target task. The choice of pretraining objective is the decisive factor: self-distillation methods (the DINO family) produce the most semantically structured representation spaces, with DINOv2 and DINOv3 at the top, the masked-image models iBOT and AttMask in the middle, and the contrastive MoCo v3 far behind. These findings confirm that modern self-supervised pretraining makes label-efficient, non-parametric classification practical today. Crucially, the strongest of these label-free backbones matches, and on CIFAR-100 exceeds, is a fully supervised ResNet-50 trained on the entire labelled dataset, underscoring the label-efficiency of modern self-supervised representations.

Appendix A. Supervised ResNet-50 Baseline Results

As a supervised reference, four ResNet-50 classifiers were trained from scratch with labels and then used as frozen feature extractors under the identical kNN protocol: CIFAR-10 and CIFAR-100, each with standard augmentation (random crop + horizontal flip) and a stronger regime (TrivialAugmentWide + random erasing). Because these features come from fully supervised training, their class clusters are already tight, so kNN accuracy is near-ceiling even at one shot. For ResNet the two pooling modes coincide (a single 2048-d vector), so only the CLS column is reported.

Values are mean % $\pm$ std % across 3 seeds at $k = 5$.

Model	Dataset / Aug	1-shot	5-shot	10-shot	100-shot
resnet50_baseline	CIFAR-10 / standard	94.4 $\pm$ 0.1	94.6 $\pm$ 0.1	94.6 $\pm$ 0.0	94.6 $\pm$ 0.1
resnet50_aug	CIFAR-10 / strong	91.3 $\pm$ 2.7	94.3 $\pm$ 0.3	94.7 $\pm$ 0.1	95.2 $\pm$ 0.2
resnet50_100	CIFAR-100 / standard	72.2 $\pm$ 0.3	75.5 $\pm$ 0.1	75.7 $\pm$ 0.1	76.6 $\pm$ 0.1
resnet50_100_aug	CIFAR-100 / strong	59.8 $\pm$ 1.2	70.8 $\pm$ 0.4	72.9 $\pm$ 0.2	76.8 $\pm$ 0.1

Label-efficiency takeaway: the best frozen self-supervised backbone matches or exceeds the supervised ResNet-50 without using any labels during pretraining. On CIFAR-10, DINOv2 reaches 94.7% versus the supervised 94.6%; on CIFAR-100, DINOv2 (80.6%) actually surpasses the supervised ResNet-50 baseline (76.6%) at 100 shots per class.

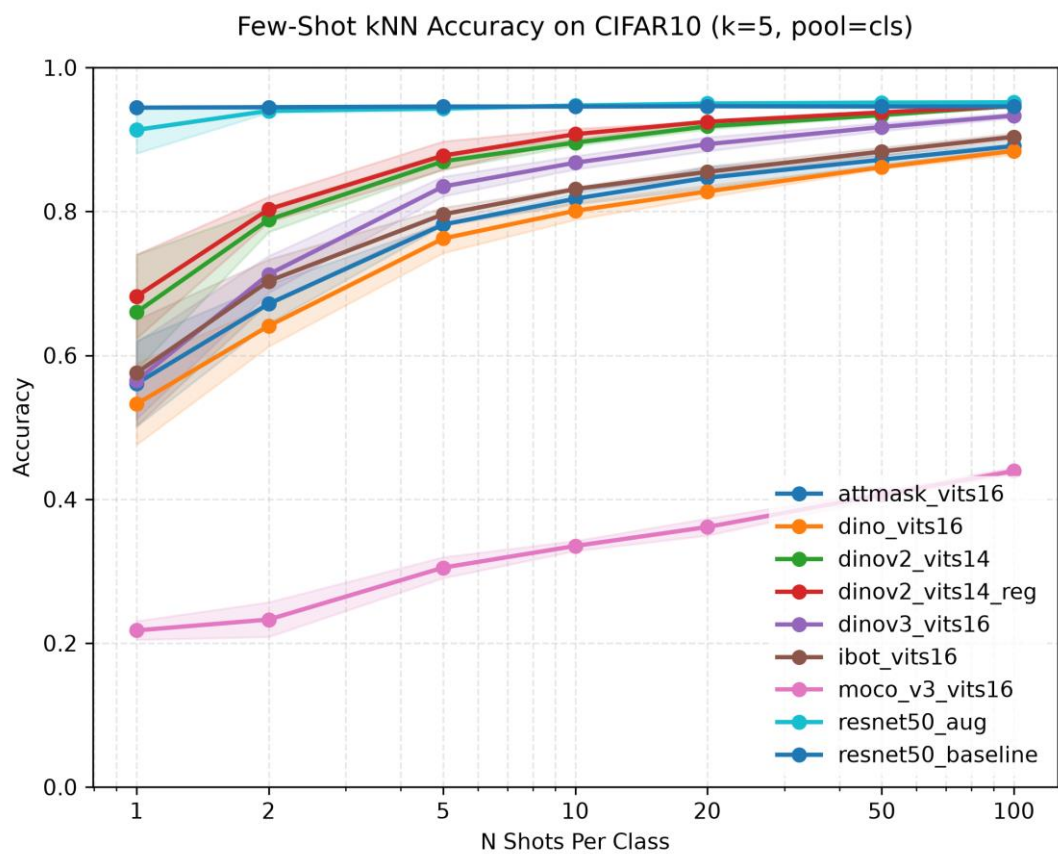


Figure A1. CIFAR-10 accuracy vs. shots (CLS, $k=5$), including the supervised ResNet-50 baselines.

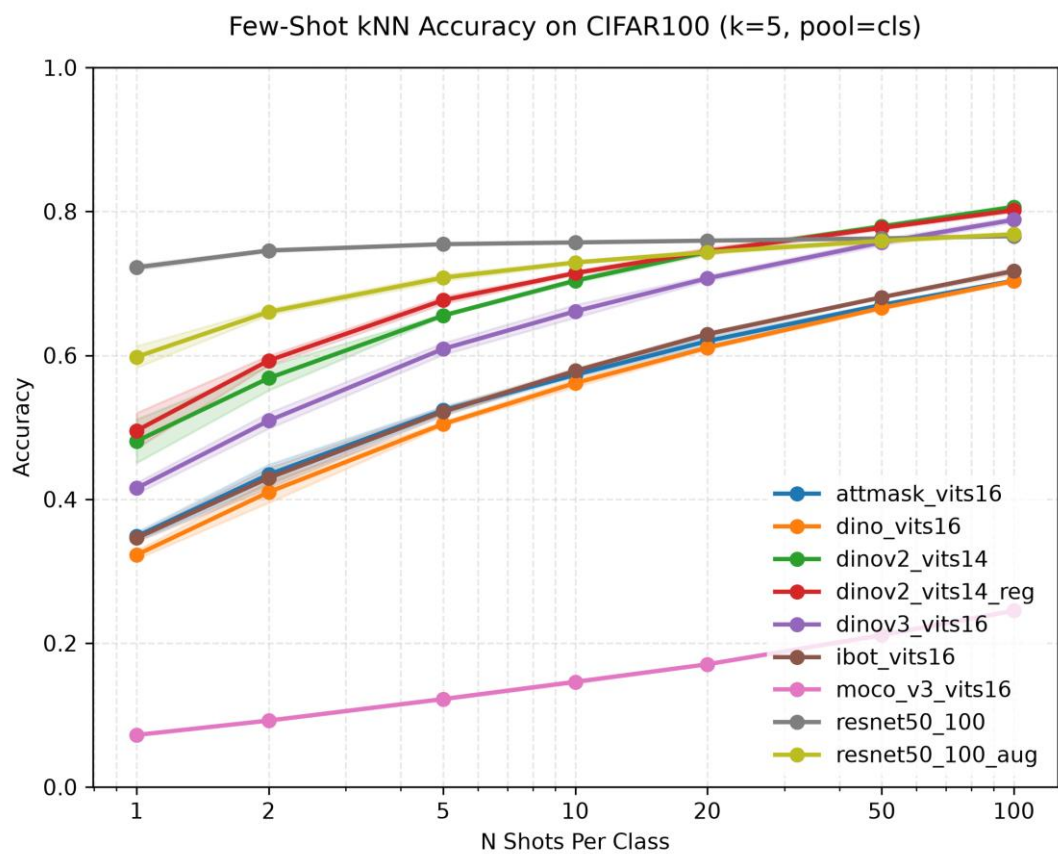


Figure A2. CIFAR-100 accuracy vs. shots (CLS, $k=5$), including the supervised ResNet-50 baselines.